

Results

01 • INSTRUMENTAL VARIABLES (IV / 2SLS)

effect with an endogenous predictor

Table 1. First stage (instrument strength)

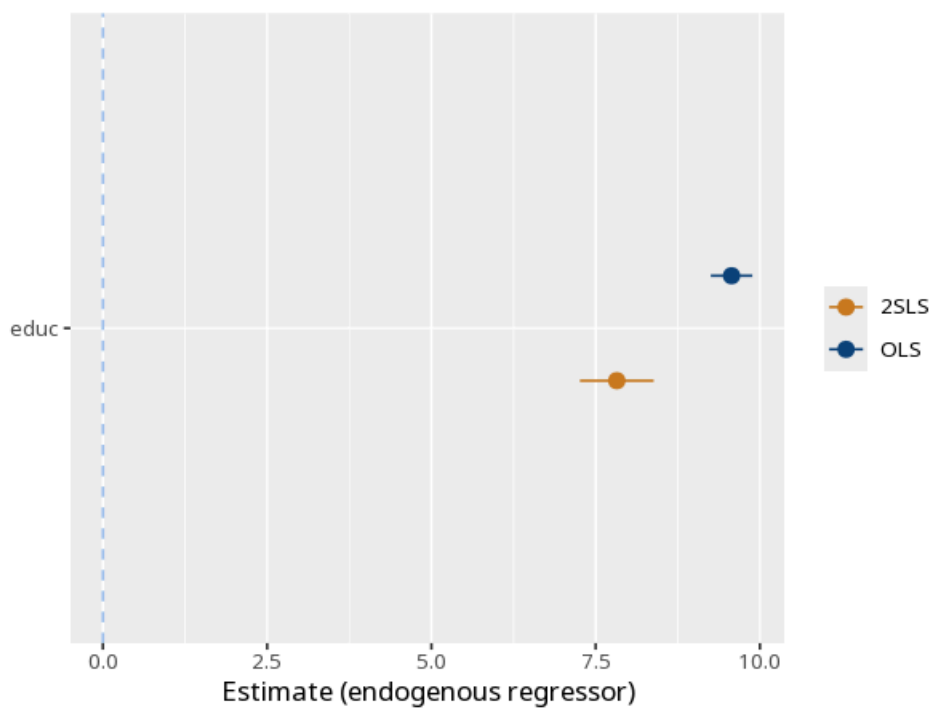
Instrument	Coef.	SE	Partial F	p
educ_iv	1.16	0.06	438.50	<.001

Table 2. 2SLS coefficients

	OLS	2SLS
(Intercept)	178.90 (2.41) [174.16, 183.65]	202.74 (4.10) [194.65, 210.83]
educ	9.57 (0.16) [9.25, 9.88]	7.82 (0.28) [7.26, 8.38]
exper	-0.02 (0.05) [-0.11, 0.08]	-0.02 (0.06) [-0.14, 0.09]
Num.Obs.	200	
RMSE	6.67	
F	373.44	
Weak-instrument F = 438.50, p < .001 (rule of thumb: F > 10)		
Wu-Hausman = 1022.75, p < .001		
Sargan: — (just-identified)		

diagnostics: weak-instrument (first-stage F), Wu-Hausman endogeneity, and Sargan over-identification (when applicable).

Figure. Coefficients



How to read this test

Check the first-stage F first (rule of thumb > 10 = strong instruments). Then read the 2SLS B for the endogenous predictor as the causal effect, with p/CI. This causal reading rests on the instrument being valid — relevant and affecting the outcome only through the endogenous predictor (the exclusion restriction). That is an assumption you must justify on theory; the Sargan test only checks over-identification (extra instruments), not validity, and is unavailable for just-identified models. Method: R ivreg / AER package; weak-instrument F (Stock & Yogo, 2005). Your significance threshold (α) is 0.05.

APA template: The 2SLS estimate for educ was $B=7.82$, $p < .001$ (first-stage $F=438.50$).

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